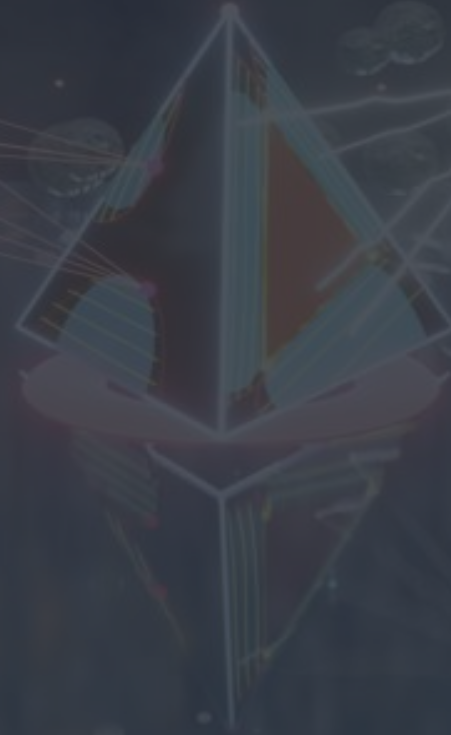
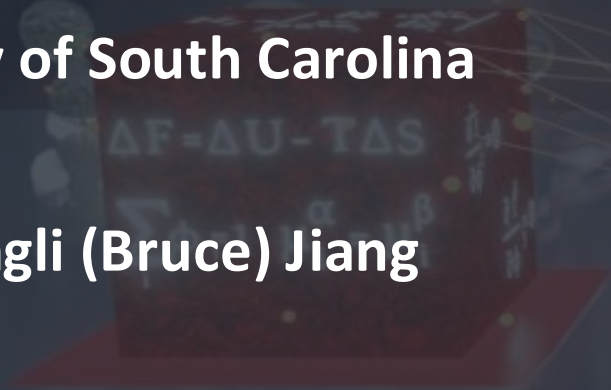


Classification II: Kernels and Decision Trees

Fall 2026

University of South Carolina

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From Lecture 13

Classification: Support Vector Machine

- *An SVM finds a maximum-margin decision boundary.*
- *Support vectors determine the position of the boundary.*
- *X controls the tradeoff between margin width and margin violations.*

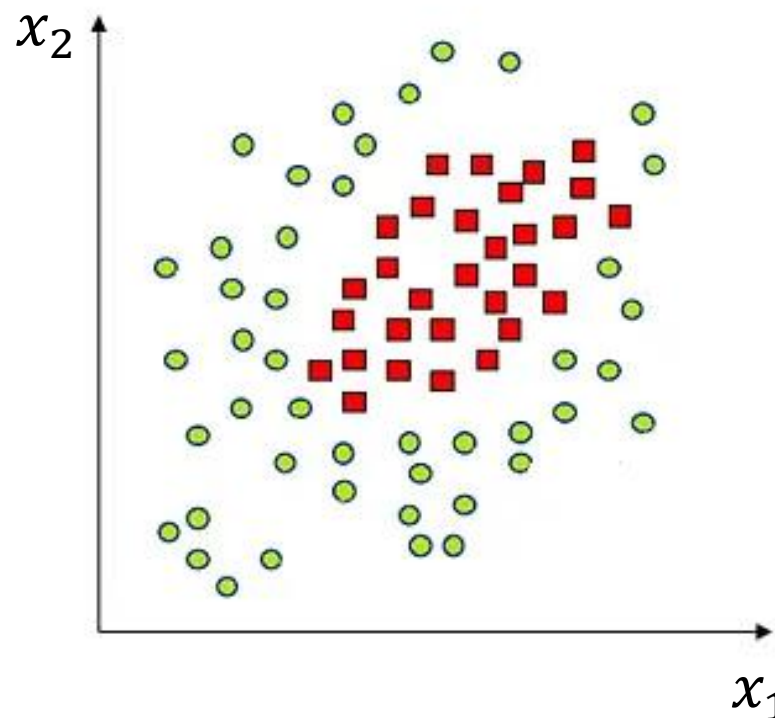
In this lecture, you will learn

- *How kernels produce nonlinear decision boundaries.*
- *How polynomial degree and RBF γ control model flexibility.*
- *How decision trees select splits using Gini impurity.*

From Nonlinear to Linear Separation

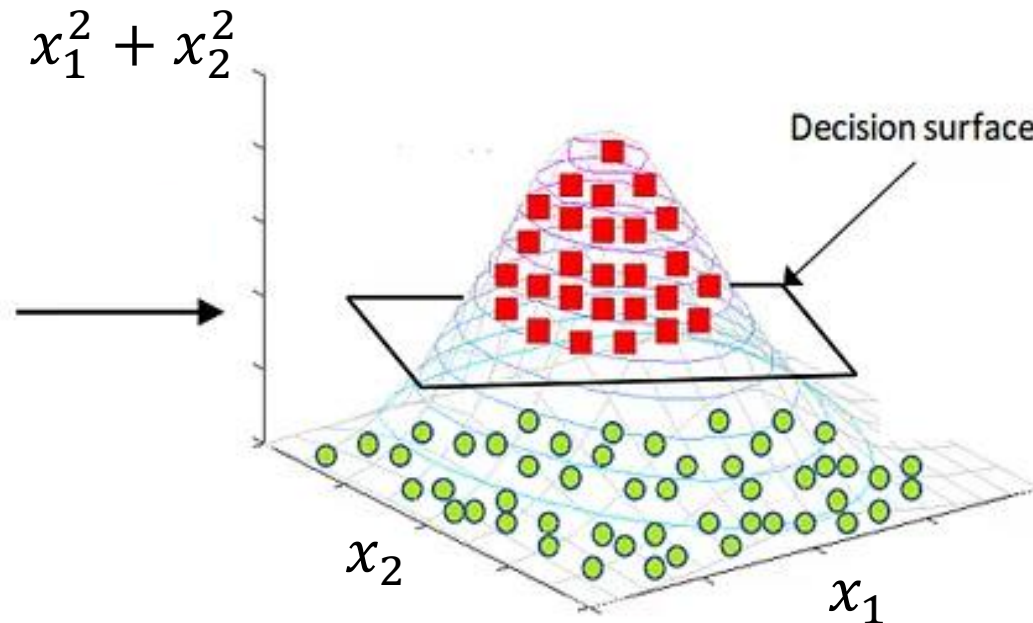
image source: [Grace Zhang](#)

Original feature space



- No straight line can separate the two classes.

Map to a richer feature space

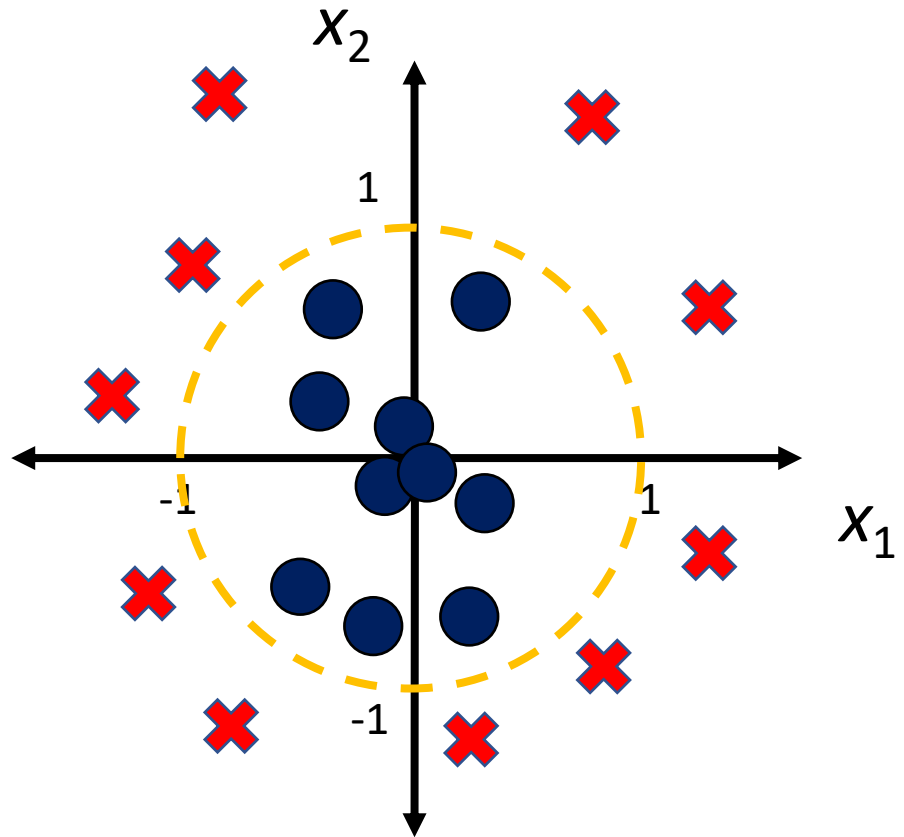


- The added feature can reveal structure that was hidden in 2D.

$$\mathbf{x} = (x_1, x_2) \rightarrow \phi(\mathbf{x}) = (x_1, x_2, x_1^2 + x_2^2)$$

SVM is still finding a linear maximum-margin separator, but in the transformed feature space.

The Kernel Trick



$$\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \longrightarrow \phi(\mathbf{x}) = \begin{bmatrix} x_1 \\ x_2 \\ x_1^2 + x_2^2 \end{bmatrix}$$

2D input space $\rightarrow$ 3D feature space

Decision boundary

$$\mathbf{w}^T \phi(\mathbf{x}) + b = 0$$

$$w_1 x_1 + w_2 x_2 + w_3 (x_1^2 + x_2^2) + b = 0$$

SVM is still finding a **linear decision boundary**.

But do we need to calculate every $\phi(x)$?

What if $\phi(x)$ has thousands or infinitely many features?

The Kernel Trick

In the dual form, SVM only needs dot products between points in the transformed feature space.

$$K(\mathbf{x}_i, \mathbf{x}_j) = \phi(\mathbf{x}_i)^T \phi(\mathbf{x}_j)$$

SVM $f(\mathbf{x}) = \mathbf{w}^T \phi(\mathbf{x}) + b$

After solving the optimization problem

$$\mathbf{w} = \sum_i \alpha_i y_i \phi(\mathbf{x}_i)$$

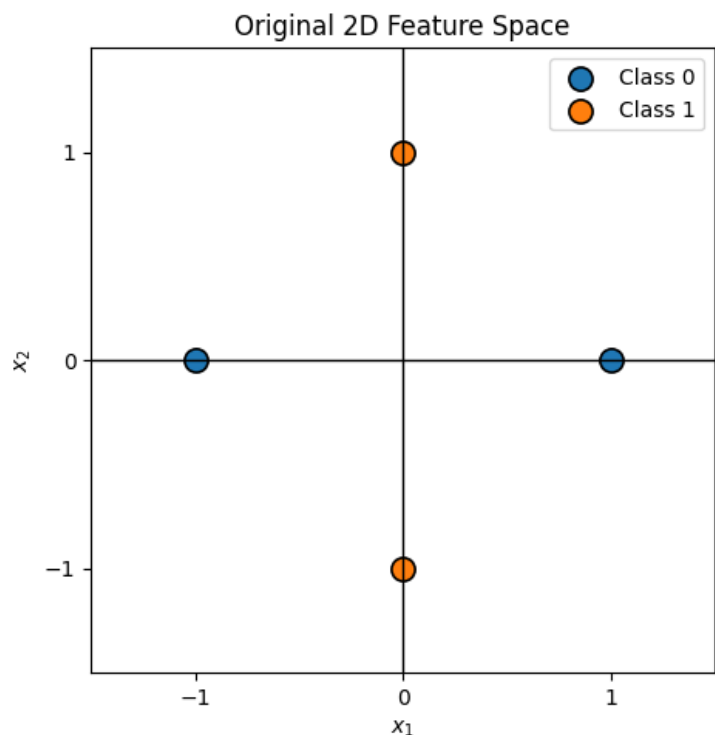
α_i 's are learned coefficients.

$$f(\mathbf{x}) = \sum_i \alpha_i y_i \underbrace{\phi(\mathbf{x}_i)^T \phi(\mathbf{x})}_{\text{dot product}} + b$$

$$f(\mathbf{x}) = \sum_i \alpha_i y_i K(\mathbf{x}_i, \mathbf{x}) + b$$

Kernel trick: compute $K(\mathbf{x}_i, \mathbf{x})$ directly, without explicitly computing $\phi(\mathbf{x})$.

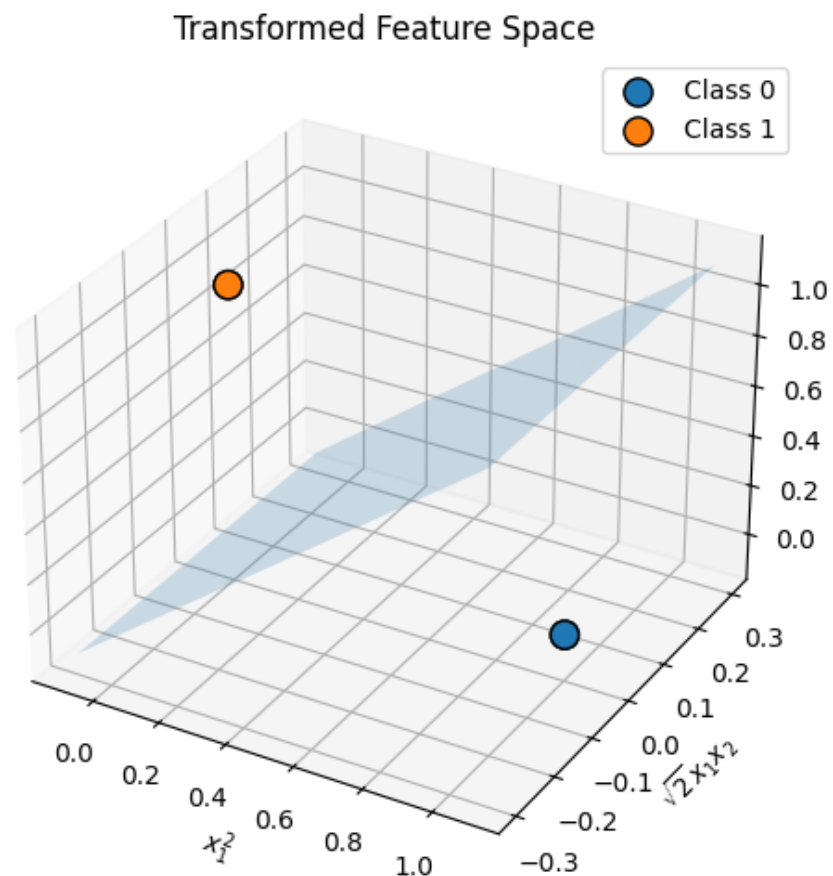
Polynomial Kernel



$$\phi(\mathbf{x}) = \begin{bmatrix} x_1^2 \\ \sqrt{2} x_1 x_2 \\ x_2^2 \end{bmatrix}$$
$$(\pm 1, 0) \rightarrow (1, 0, 0)$$
$$(0, \pm 1) \rightarrow (0, 0, 1)$$

$\mathbf{z}$ is a new input

$$\begin{aligned} \phi(\mathbf{x})^T \phi(\mathbf{z}) &= x_1^2 z_1^2 + 2x_1 x_2 z_1 z_2 + x_2^2 z_2^2 \\ &= (x_1 z_1 + x_2 z_2)^2 \\ &= (\mathbf{x}^T \mathbf{z})^2 \end{aligned}$$



**Linearly separable
in feature space**

We get the dot product
in the 3D feature space
without ever calculating
the 3D coordinates.

Kernel

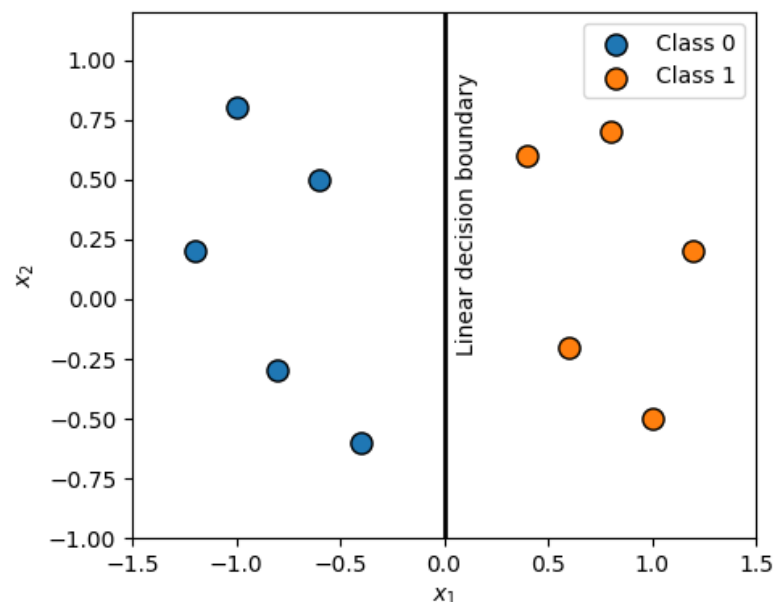
$$K(\mathbf{x}, \mathbf{z}) = (\mathbf{x}^T \mathbf{z})^2$$

What Does a Kernel Actually Do?

$$K(\mathbf{x}, \mathbf{z}) = \phi(\mathbf{x})^T \phi(\mathbf{z})$$

A kernel measures similarity between two samples in the feature space.

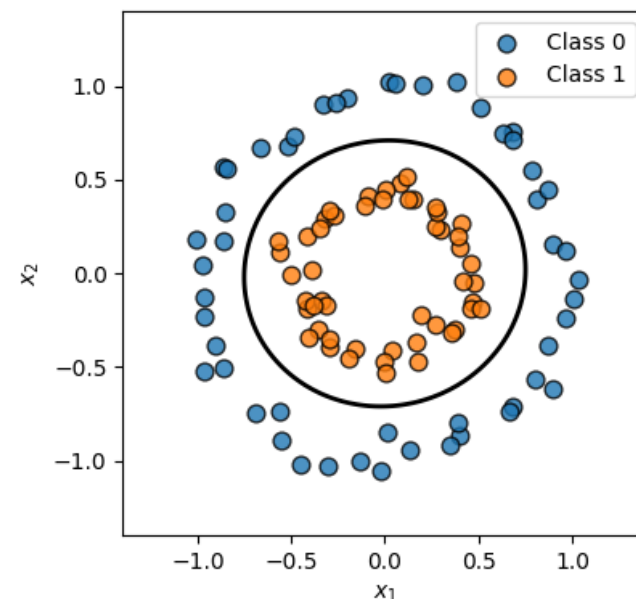
Linear kernel $K(\mathbf{x}, \mathbf{z}) = \mathbf{x}^T \mathbf{z}$



No nonlinear transformation

- Similar direction $\rightarrow$ larger dot product
- Produces a linear boundary

Polynomial kernel $K(\mathbf{x}, \mathbf{z}) = (\mathbf{x}^T \mathbf{z})^d$



Nonlinear transformation

- Adds interactions and polynomial features
- Higher $d \rightarrow$ more complex feature interactions

Radial Basis Function (RBF) kernel

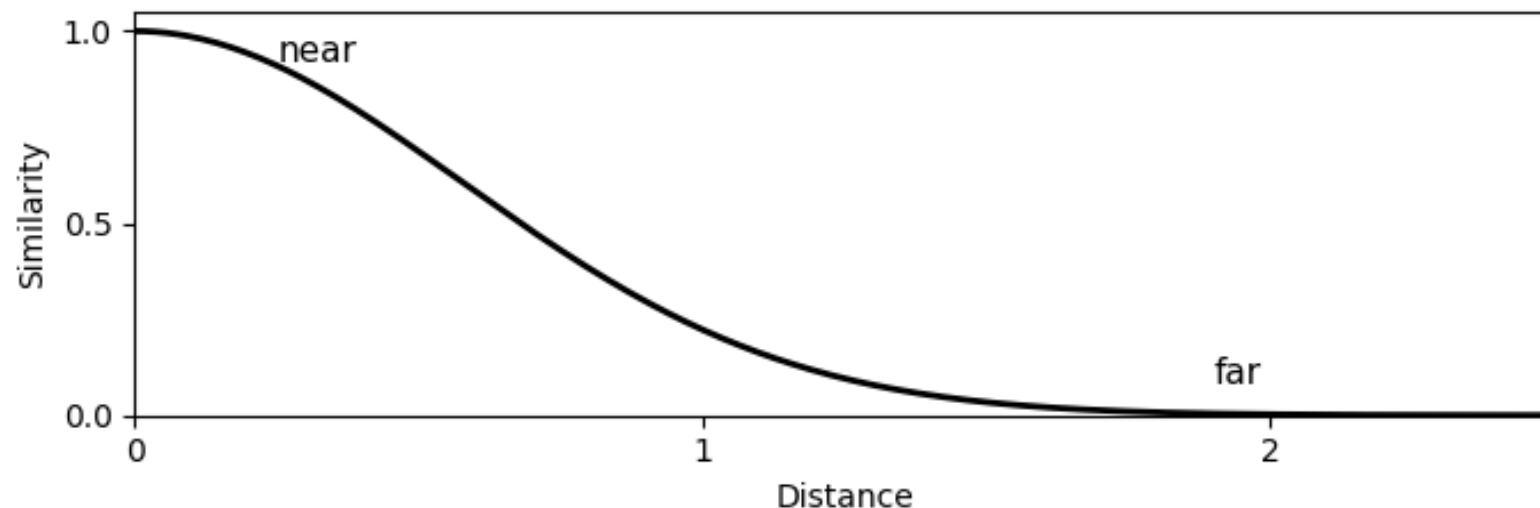
$$K(\mathbf{x}, \mathbf{z}) = \exp(-\gamma \|\mathbf{x} - \mathbf{z}\|^2)$$

Nearby points strongly influence each other; distant points have little influence.

$$\mathbf{x} \approx \mathbf{z} \quad \Rightarrow \quad K \approx 1$$

$$\mathbf{x} \text{ somewhat far from } \mathbf{z} \quad \Rightarrow \quad 0 < K < 1$$

$$\mathbf{x} \text{ very far from } \mathbf{z} \quad \Rightarrow \quad K \approx 0$$



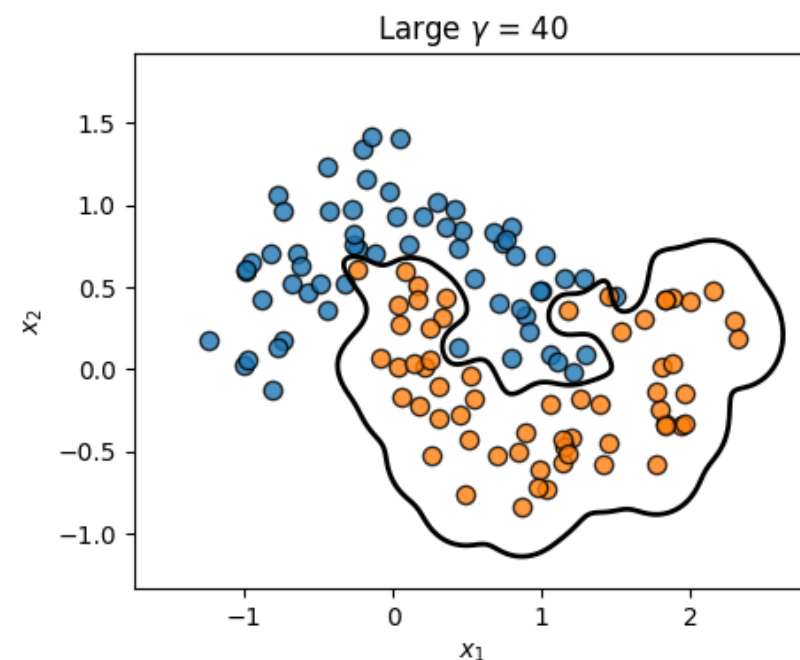
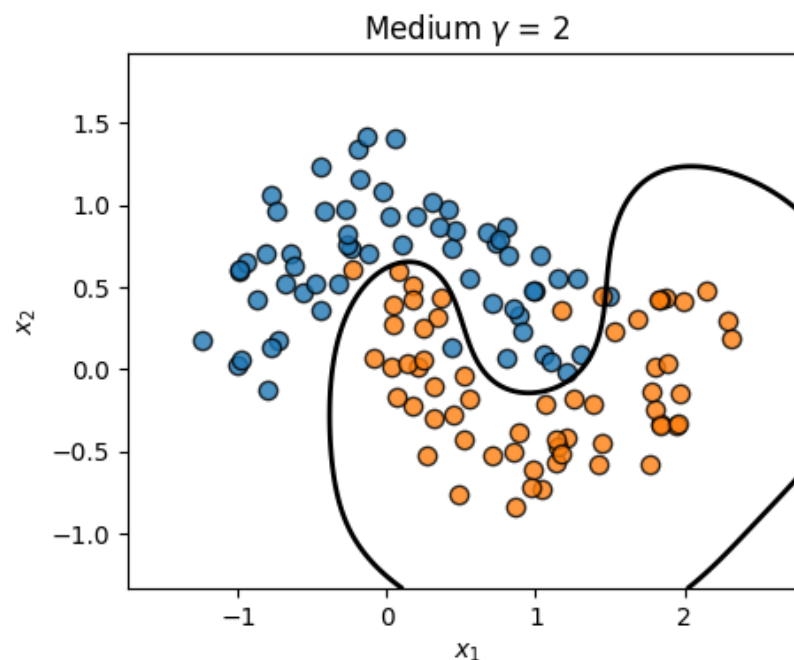
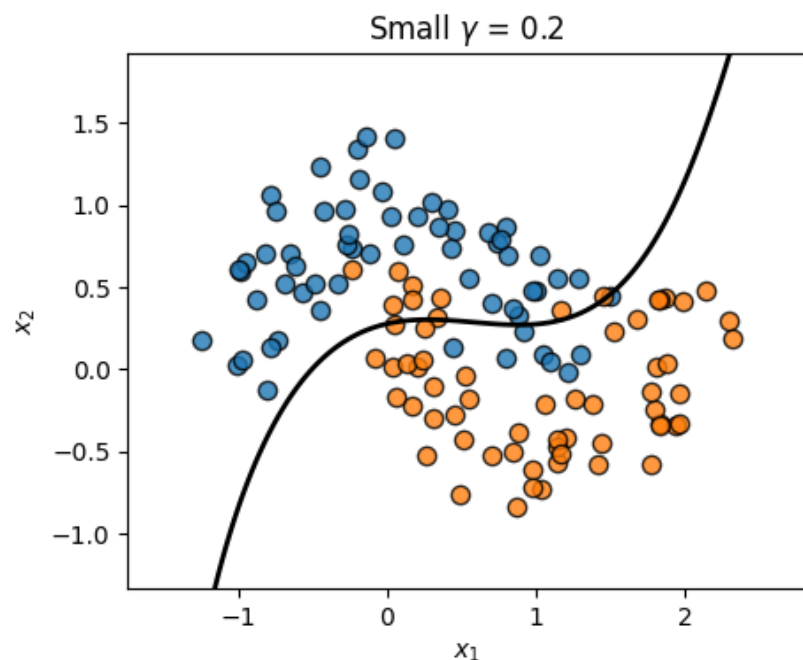
Taylor expansion

$$k(\mathbf{x}, \mathbf{z}) = C \left(\sum_{n=0}^{\infty} \frac{(\mathbf{x}^T \mathbf{z})^n}{n!} \right)$$

Infinite dimension
dot product

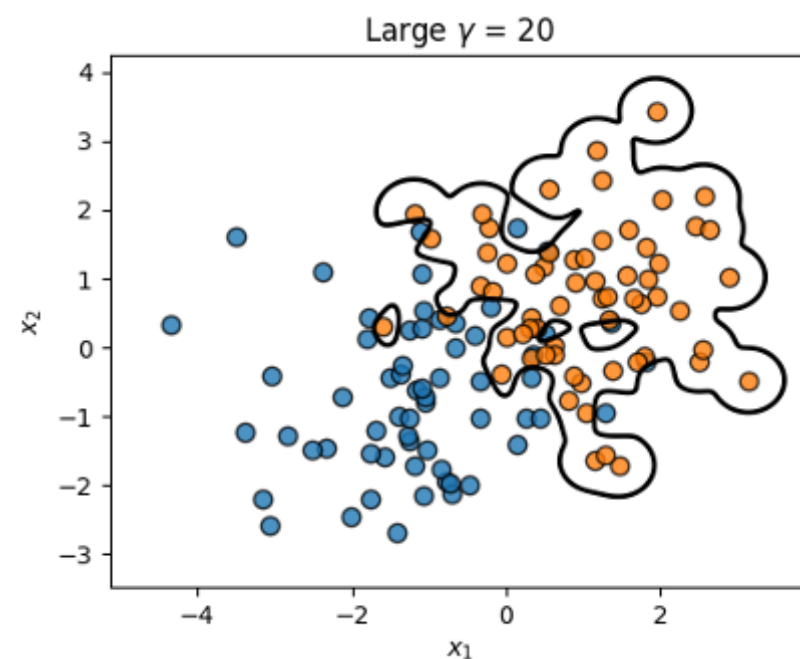
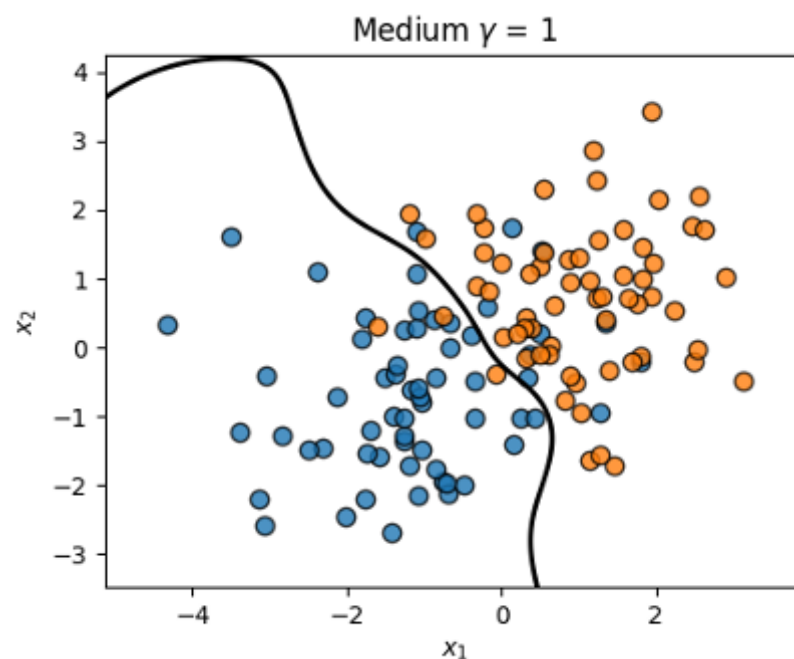
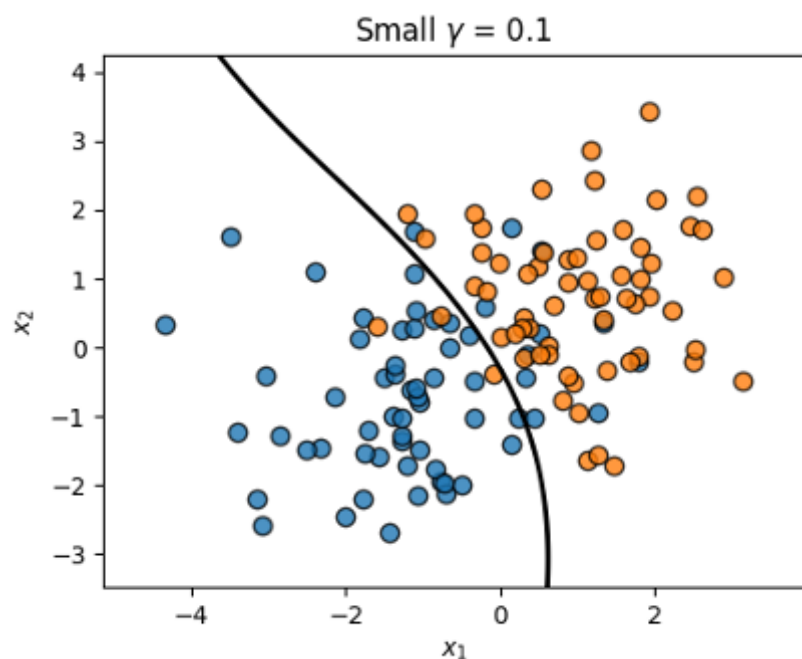
RBF Kernel: What does γ do?

- **small** $\gamma \rightarrow$ broad influence $\rightarrow$ smoother boundary
- **medium** $\gamma \rightarrow$ reasonable boundary
- **large** $\gamma \rightarrow$ very local influence $\rightarrow$ wiggly / possible overfitting



RBF Kernel: What does γ do?

- **small** $\gamma \rightarrow$ broad influence $\rightarrow$ smoother boundary
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Case Study: Breast Cancer Classification

Goal: predict whether a tumor is **malignant** or **benign** from measured cell/nucleus features.

- 569 samples
- 30 numerical features
- Binary classification
- Examples: radius, texture, perimeter, smoothness
- We visualize only 2 features below; the SVM will use **all 30**.

```
import pandas as pd
import matplotlib.pyplot as plt
from sklearn.datasets import load_breast_cancer

data = load_breast_cancer(as_frame=True)

X = data.data
y = data.target

print(X.shape)
print(data.target_names)
X.head()

(569, 30)
['malignant' 'benign']
```

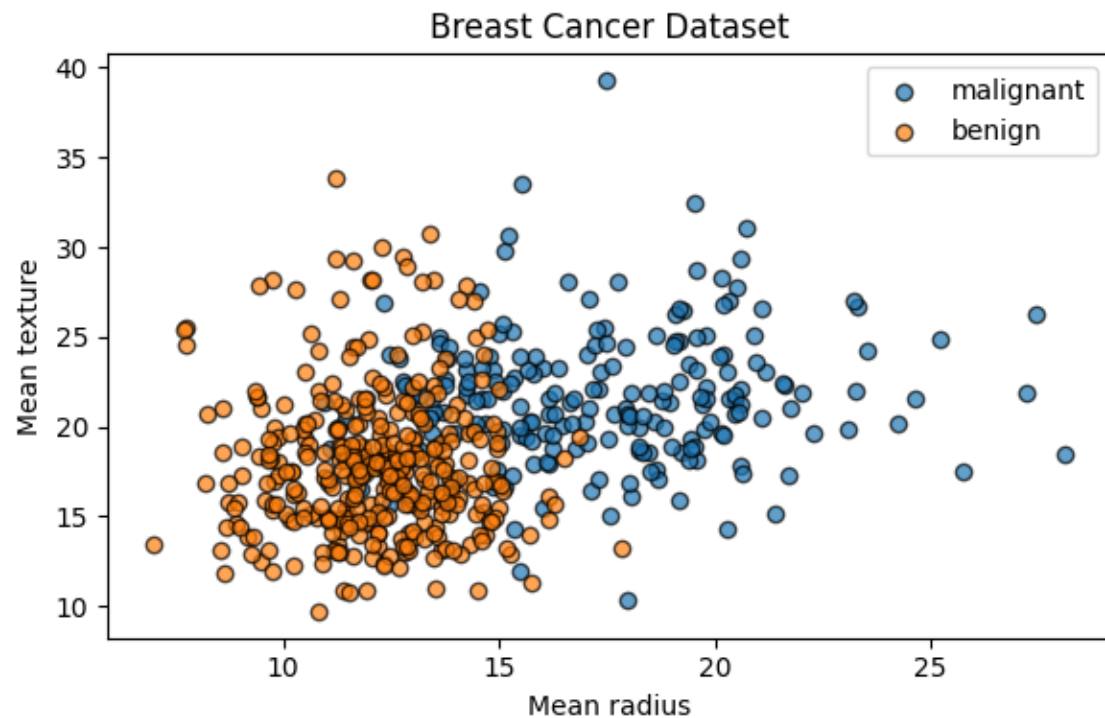
1. Load the data

	mean radius	mean texture	mean perimeter	mean area	mean smoothness	mean compactness	mean concavity
0	17.99	10.38	122.80	1001.0	0.11840	0.27760	0.3001
1	20.57	17.77	132.90	1326.0	0.08474	0.07864	0.0869
2	19.69	21.25	130.00	1203.0	0.10960	0.15990	0.1974
3	11.42	20.38	77.58	386.1	0.14250	0.28390	0.2414
4	20.29	14.34	135.10	1297.0	0.10030	0.13280	0.1980

5 rows × 30 columns

Case Study: Breast Cancer Classification

2. Visualize two features



- There is some separation, but also substantial overlap.
- We have 30 features, so the true classification problem is much higher-dimensional than this plot.

3. Train / test split

```
from sklearn.model_selection import train_test_split

X_train, X_test, y_train, y_test = train_test_split(
    X, y,
    test_size=0.2,
    stratify=y,
    random_state=26
)

print(len(X_train), len(X_test))

455 114
```

- `stratify=y`: Keep approximately the same malignant/benign proportion in each split.

Case Study: Breast Cancer Classification

4. Build the pipeline

```
from sklearn.pipeline import Pipeline
from sklearn.preprocessing import StandardScaler
from sklearn.svm import SVC

pipe = Pipeline([
    ("scaler", StandardScaler()),
    ("svm", SVC())
])
```

- SVM depends on distances and dot products, so feature scaling is important.

Kernel	Tune
Linear	C
Polynomial	C, d
RBF	C, γ

5. Define hyperparameters

```
param_grid = [
    {
        "svm__kernel": ["linear"],
        "svm__C": [0.1, 1, 10]
    },
    {
        "svm__kernel": ["poly"],
        "svm__C": [0.1, 1, 10],
        "svm__degree": [2, 3, 4]
    },
    {
        "svm__kernel": ["rbf"],
        "svm__C": [0.1, 1, 10],
        "svm__gamma": [0.001, 0.01, 0.1, 1]
    }
]
```

Case Study: Breast Cancer Classification

6. 5-fold CV for hyperparameter tuning

```
from sklearn.model_selection import GridSearchCV
from sklearn.metrics import accuracy_score

search = GridSearchCV(
    pipe,
    param_grid,
    cv=5,
    scoring="accuracy"
)

search.fit(X_train, y_train)

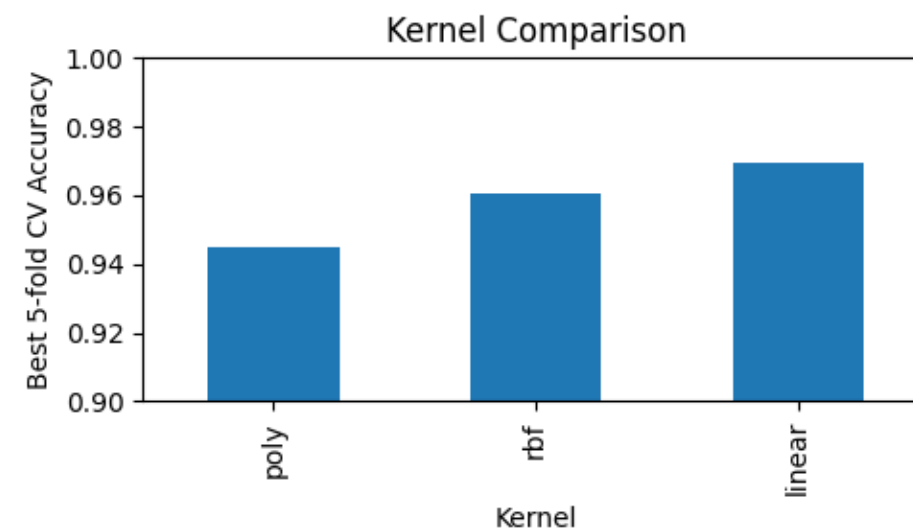
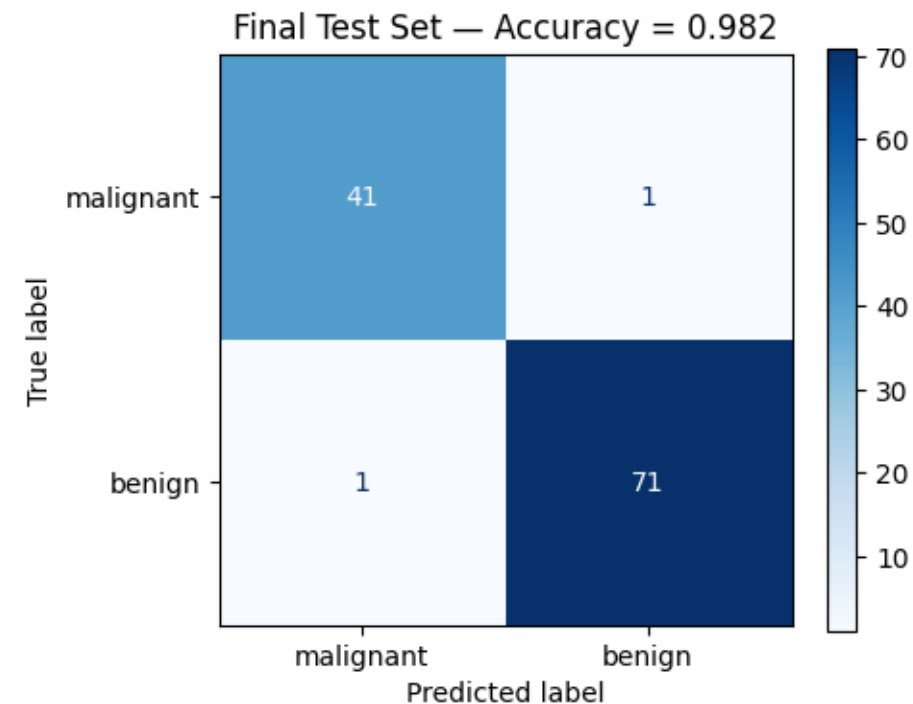
print("Best CV accuracy:", search.best_score_)
print("Best parameters:", search.best_params_)

Best CV accuracy: 0.9692307692307693
Best parameters: {'svm__C': 10, 'svm__kernel': 'linear'}
```

```
test_acc = accuracy_score(
    y_test,
    search.predict(X_test)
)

print("Test accuracy:", test_acc)

Test accuracy: 0.9824561403508771
```

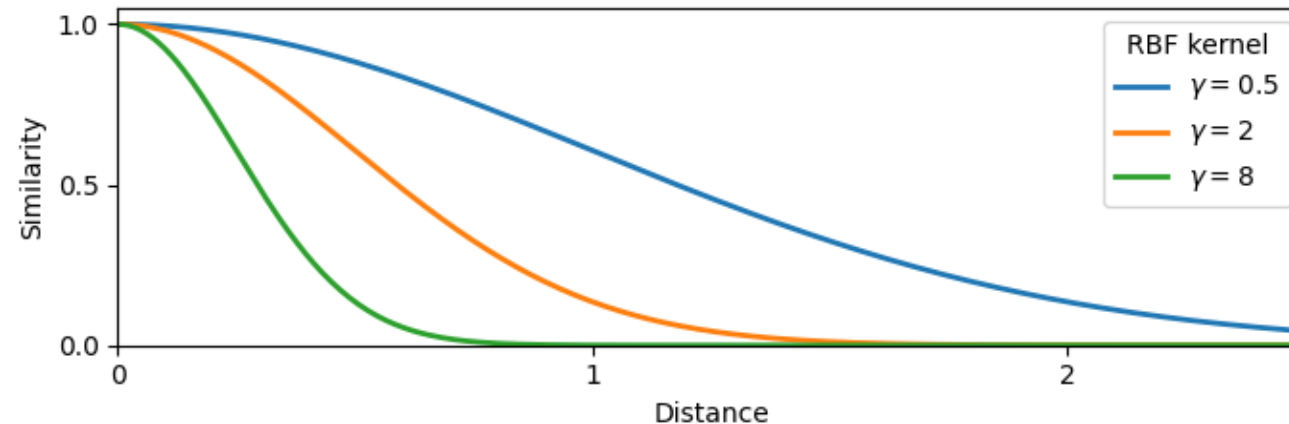


1. For an RBF SVM, what is the main effect of increasing γ while keeping C fixed?

- A. The penalty for misclassification becomes larger
- B. The polynomial degree of the decision boundary increases
- C. Each training point has a more local influence, allowing a more flexible boundary
- D. Each training point has a broader influence, producing a smoother boundary

Answer: C. Increasing γ makes the RBF similarity decay more rapidly with distance, so each point influences a smaller neighborhood and the boundary can become more complex.

1. For an RBF SVM, what is the main effect of increasing γ while keeping C fixed?
- A. The penalty for misclassification becomes larger
 - B. The polynomial degree of the decision boundary increases
 - C. Each training point has a more local influence, allowing a more flexible boundary
 - D. Each training point has a broader influence, producing a smoother boundary



2. Which hyperparameters would you naturally tune when comparing linear, polynomial, and RBF SVM kernels?

- A. Linear: degree; Polynomial: γ ; RBF: C only
- B. Linear: C and γ ; Polynomial: degree only; RBF: C and degree
- C. Linear: C; Polynomial: C and degree; RBF: C and γ
- D. Linear: γ ; Polynomial: C; RBF: degree

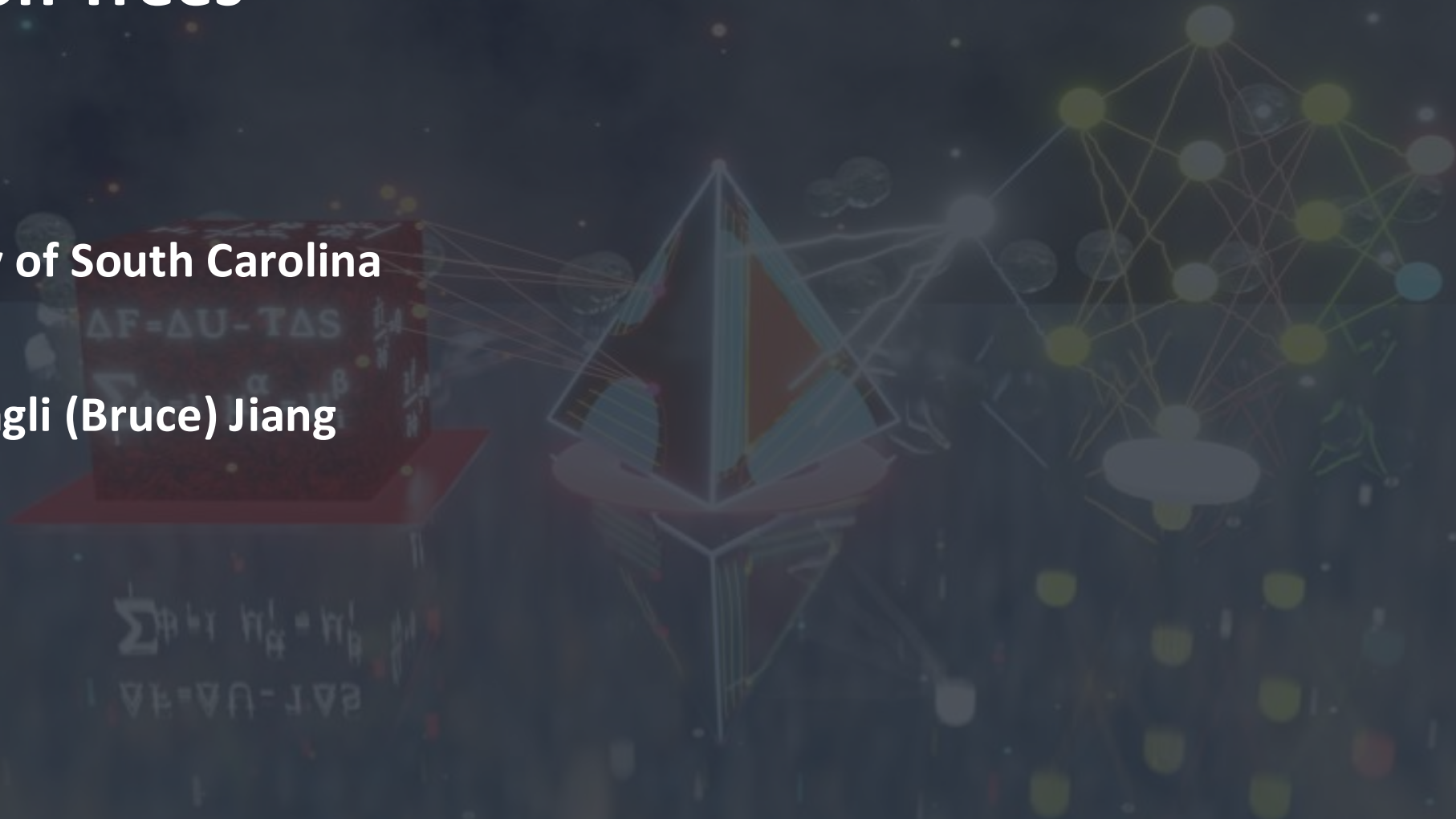
Answer: C.

Decision Trees

Fall 2026

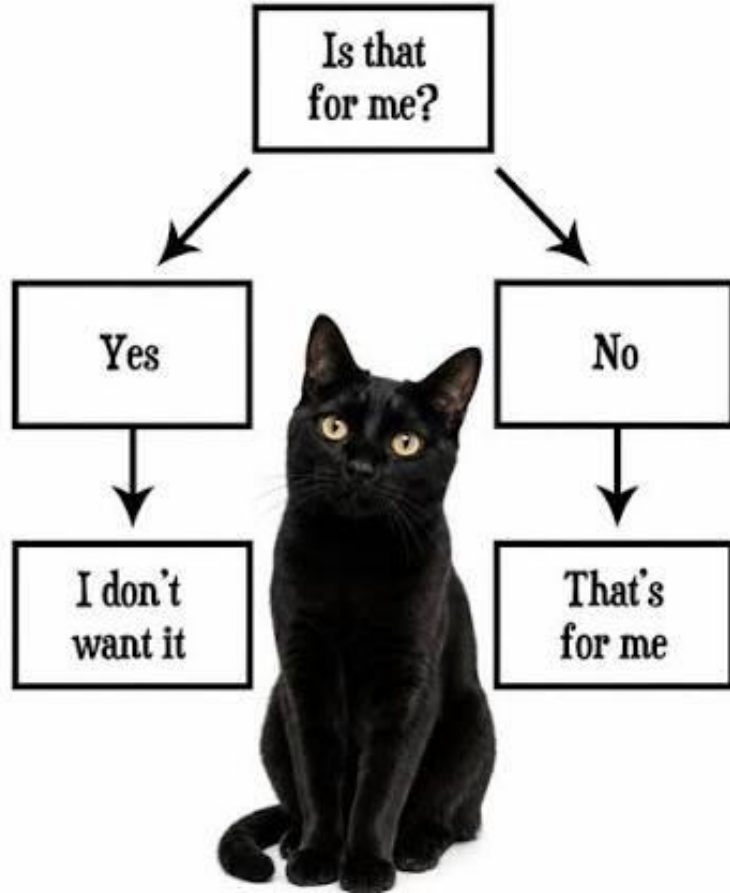
University of South Carolina

Prof. Shengli (Bruce) Jiang

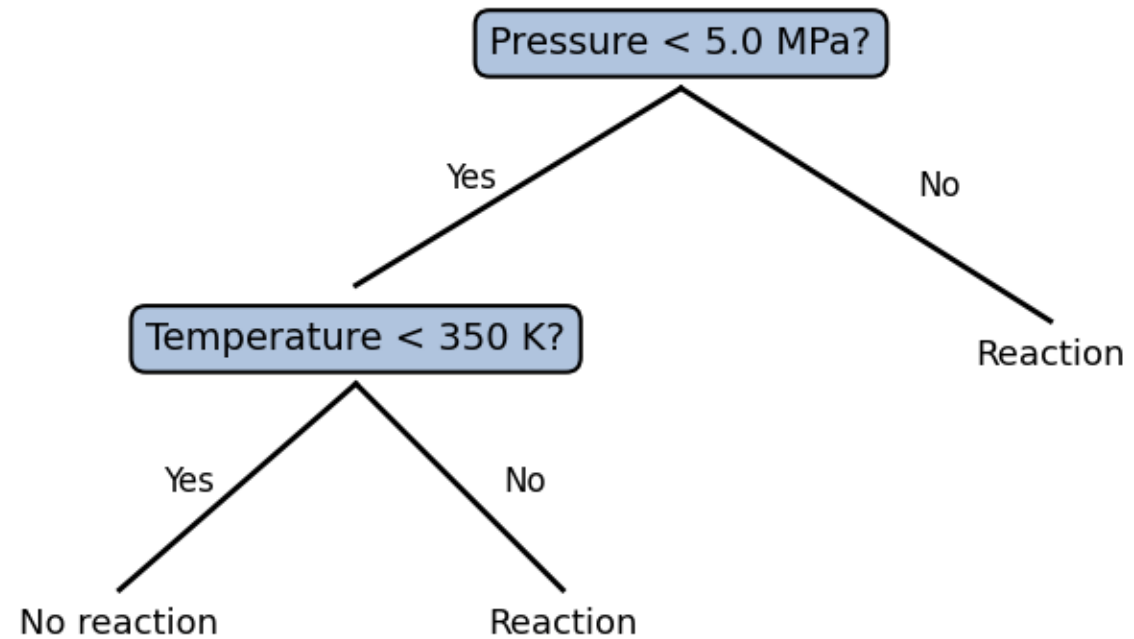


Decision Tree

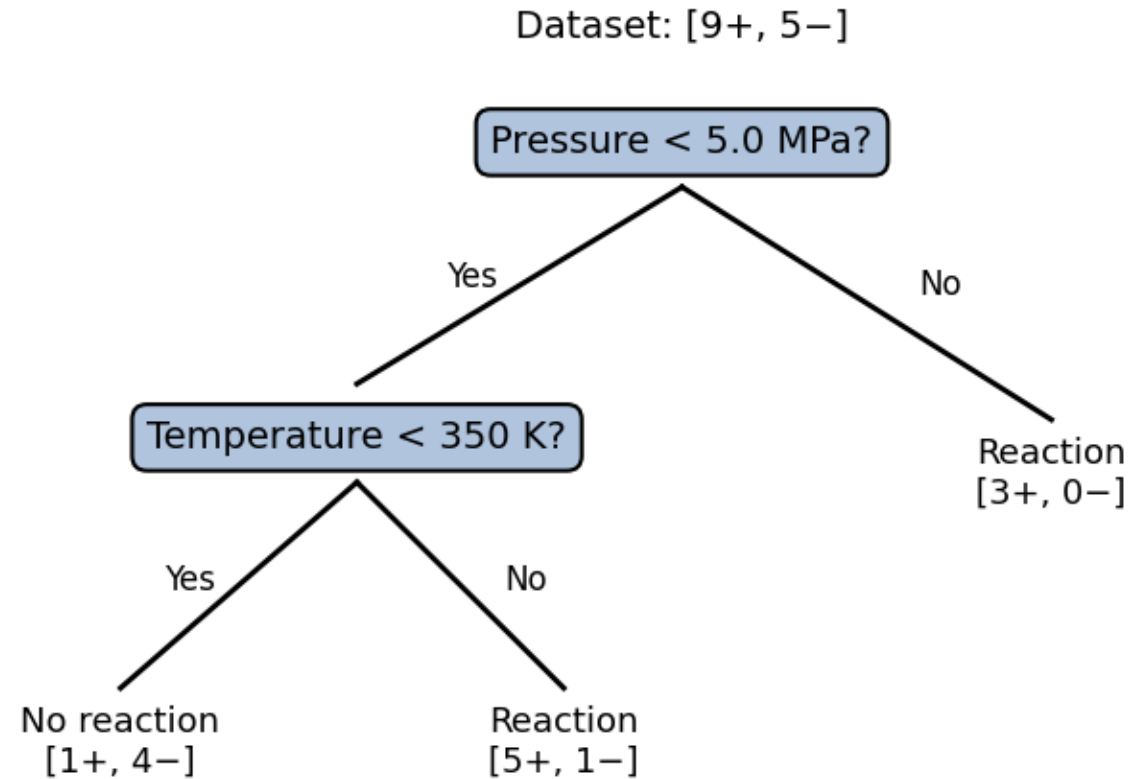
Cat's Decision Making Tree



A decision tree partitions the feature space using simple if-then rules.



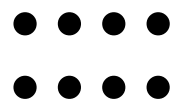
How Does a Tree Split the Data?



- The algorithm considers many possible features and thresholds.
 - Pressure < 2.0 Mpa, Pressure < 3.0 Mpa, Temperature < 340 K
- **Which split makes the resulting groups as pure as possible?**

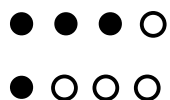
What is a Good Split?

Node A



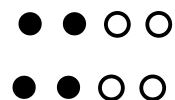
pure

Node B



mixed

Node C



very mixed

For a dataset D with k classes, the **Gini impurity** is

$$G = \text{Gini}(D) = 1 - \sum_{i=1}^k p_i^2$$

For binary classification:

$$G = 1 - (p_0^2 + p_1^2)$$

Pure node:

$$p_0 = 1, p_1 = 0$$

$$G = 1 - (1^2 + 0^2) = 0$$

50/50 node:

$$p_0 = 0.5, p_1 = 0.5$$

$$G = 1 - (0.5^2 + 0.5^2) = 0.5$$

Lower impurity = cleaner separation of classes.

Finding the Best Split

Parent

● ● ● ○ ○ ○

Candidate split A:

Left

Right

● ● ○

● ○ ○

Candidate split B:

Left

Right

● ● ●

○ ○ ○

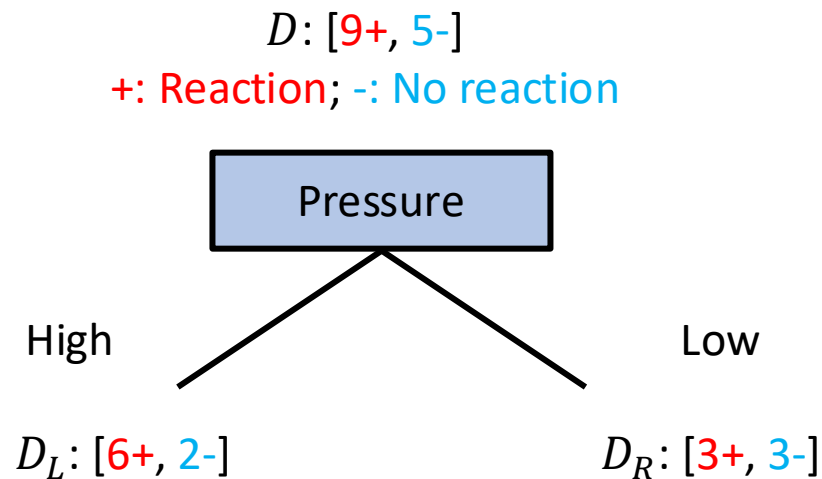
$$G_{\text{split}} = \text{weighted impurity after split} = \frac{N_L}{N} G_L + \frac{N_R}{N} G_R$$

- $|D|$: number of samples in the parent node
- $|D_L|$: number of samples that go to the **left child**
- $|D_R|$: number of samples that go to the **right child**
- $|D| = |D_L| + |D_R|$

Choose the split with the largest impurity reduction.

Which split is better?

Example



$$\text{Gini}(D_L) = 1 - \left(\frac{6}{8}\right)^2 - \left(\frac{2}{8}\right)^2 = 0.375$$

$$\text{Gini}(D) = 1 - \sum_{i=1}^k p_i^2$$

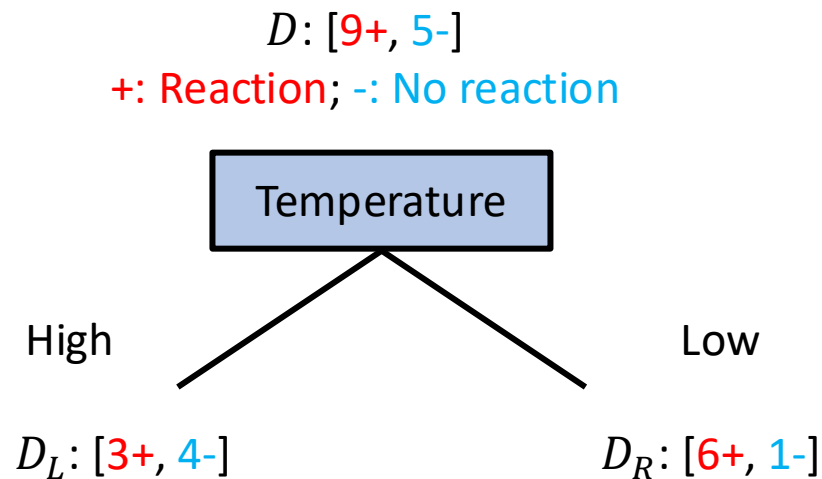
$$\text{Gini}_{\text{split}}(D) = \frac{|D_L|}{|D|} \text{Gini}(D_L) + \frac{|D_R|}{|D|} \text{Gini}(D_R)$$

$$\text{Gini}(D) = 1 - \left(\frac{9}{14}\right)^2 - \left(\frac{5}{14}\right)^2 = 0.459$$

$$\text{Gini}(D_R) = 1 - \left(\frac{3}{6}\right)^2 - \left(\frac{3}{6}\right)^2 = 0.5$$

$$\text{Gini}_{\text{split}}(D) = \frac{8}{14} \cdot 0.375 + \frac{6}{14} \cdot 0.5 = 0.429$$

Example



$$\text{Gini}(D_L) = 1 - \left(\frac{3}{7}\right)^2 - \left(\frac{4}{7}\right)^2 = 0.490$$

$$\text{Gini}(D) = 1 - \sum_{i=1}^k p_i^2$$

$$\text{Gini}_{\text{split}}(D) = \frac{|D_L|}{|D|} \text{Gini}(D_L) + \frac{|D_R|}{|D|} \text{Gini}(D_R)$$

$$\text{Gini}(D) = 1 - \left(\frac{9}{14}\right)^2 - \left(\frac{5}{14}\right)^2 = 0.459$$

$$\text{Gini}(D_R) = 1 - \left(\frac{6}{7}\right)^2 - \left(\frac{1}{7}\right)^2 = 0.245$$

$$\text{Gini}_{\text{split}}(D) = \frac{7}{14} \cdot 0.490 + \frac{7}{14} \cdot 0.245 = 0.368$$

Is it better to split on pressure or temperature?

1. When building a classification decision tree, what makes one candidate split better than another?

- A. It creates child nodes with lower weighted impurity
- B. It always uses the feature with the largest numerical values
- C. It creates the largest possible number of branches
- D. It creates the same number of samples in both child nodes

Answer: A. A classification tree prefers the split that makes the resulting child nodes purer overall, equivalently giving the largest reduction in impurity.

Interactive Python Time

`14_kernel.ipynb`